

(TR-104) PYTHON WITH AI/ML

Training Day 28 Report

Date: 18 February 2026

Topic Covered: Resume Category Classification, Model Comparison & Real-World NLP Improvements

The twenty-eighth day of training focused on building and improving a supervised NLP-based Resume Classification system. The work included handling real-world resume edge cases, comparing multiple machine learning models, improving preprocessing techniques, and performing detailed error analysis.

- Built a Resume Category Classification model for predicting resume domains such as Data Science, Web Development, HR, and Marketing.
- Understood the difference between similarity-based retrieval systems and supervised text classification.
- Loaded and structured resume datasets for NLP model training.
- Performed advanced text cleaning including removal of emails, phone numbers, symbols, and formatting noise.
- Applied preprocessing techniques such as tokenization, stopword removal, and lemmatization.
- Compared Bag of Words, TF-IDF, and TF-IDF with bigrams for feature extraction.
- Implemented TF-IDF vectorization with parameter tuning for improved model performance.
- Used Train, Validation, and Test split methodology for better evaluation and unbiased testing.
- Trained and compared Naive Bayes and Logistic Regression models.
- Evaluated models using accuracy, precision, recall, and F1-score metrics.
- Generated confusion matrix to analyze overlapping categories and weak predictions.
- Performed error analysis on incorrect predictions and identified reasons behind model confusion.

- Analyzed resumes containing mixed-domain skills and overlapping keywords.
- Added prediction confidence scores for category prediction outputs.
- Strengthened understanding of real-world NLP pipeline improvement and classification challenges.
- **GitHub Link:**
https://github.com/JaspinderKaurWalia26/REAL_RESUME_CLASSIFIER
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/building-a-smarter-resume-classifier-logistic-regression-naive-bayes-ad2525292205>